

while Loop (10 Problems)

1. Reverse a number

```
#include <iostream>
using namespace std;

// Reverse a number
int main(){
    int num;
    cout<<"Enter a Number: ";
    cin >> num;

    //One method
    int rev = 0;
    while (num > 0){
        rev = (10 * rev) + num % 10;
        num = num /10;
    }
    cout <<rev;

    //Aother Method
    int digit = 0;
    while(num > 0){
        digit = num % 10;
        num = num / 10;
        if (digit < 9){
            cout <<digit <<" ";
        }
    }

    return 0;
}
```

2. Count digits in a number

```
#include <iostream>
using namespace std;

// Count digits in a number
int main()
{
    int num;
    cout << "Enter a Number: ";
    cin >> num;

    int count = 0;
```

```
while (num > 0)
{
    num = num / 10;
    count++;
}
cout << count;

return 0;
}
```

3. Sum digits of a number

```
#include <iostream>
using namespace std;

// Sum digits of a number
int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >>num;

    int sum = 0;
    while(num > 0){
        sum = sum + (num % 10);
        num = num /10;
    }
    cout <<"Sum Of digits in Number: " <<sum <<endl; ;

    return 0;
}
```

4. Check palindrome number

```
#include <iostream>
using namespace std;

// Check palindrome number
int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >> num;

    int original_num = num;
    int rev = 0;
    while(num > 0){
        rev = (rev * 10) + (num % 10);
        num = num / 10;
    }
    // cout <<rev;
```

```
if (original_num == rev){
    cout <<original_num <<" this is Palindrome Number." <<endl;
}else{
    cout <<"Not Palindrome" <<endl;
}

return 0;
}
```

5. Print digits of a number one by one

```
#include <iostream>
using namespace std;

// Print digits of a number one by one
int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >> num;

    int rev = 0;
    int temp = num;

    // Reverse the number
    while (temp > 0) {
        rev = (rev * 10) + (temp % 10);
        temp = temp / 10;
    }

    // Print digits in original order
    while (rev > 0) {
        cout << rev % 10 << " ";
        rev = rev / 10;
    }

    return 0;
}
```

6. Find the product of digits

```
#include <iostream>
using namespace std;

// Find the product of digits
int main(){
    int num;
    cout <<"Enter a number: ";
    cin >> num;
```

```
int mul = 1;

while(num > 0){
    mul = mul * (num % 10);
    num = num /10;
}
cout <<mul <<" is product of digits"<<endl;

return 0;
}
```

7. Find the largest digit in a number

```
#include <iostream>
using namespace std;

//Find the largest digit in a number
int main(){
    int num;
    cout <<"Enter a Number: ";
    cin >>num;
    int largest_digit = 0;
    int digit;

    while(num > 0){
        digit = num % 10;
        if (digit > largest_digit){
            largest_digit = digit;
        }
        num = num / 10;
    }
    cout <<"Largest Digit " <<largest_digit <<endl;

    return 0;
}
```

8. Convert decimal to binary

```
#include <iostream>
using namespace std;

// Convert Decimal to Binary
int main() {
    int num;
    cout << "Enter a Decimal Number: ";
    cin >> num;
```

```
int binary[32];
int i = 0;

while (num > 0) {
    binary[i] = num % 2;
    num = num / 2;
    i++;
}

cout << "Binary = ";

while (i > 0) {
    i--;
    cout << binary[i];
}

return 0;
}
```

9. Calculate power (e.g., 2^5)

```
#include <iostream>
using namespace std;

int main(){
    int base,exp;
    cout <<"Enter base: ";
    cin >> base;
    cout <<"Enter exponential: ";
    cin >> exp;

    int power = 1;
    int i = 1;

    while(i <= exp){
        power = power * base;
        i++;
    }
    cout <<base <<" Power of " <<exp <<" is " <<power <<endl;
    return 0;
}
```

10. Keep taking input until the user enters 0

```
#include <iostream>
using namespace std;
```

```
//Keep taking input until the user enters 0
int main(){
    int num;

    while(true){
        cout <<"Enter number(0 to STOP):- " ;
        cin >>num;

        if(num == 0){
            break;
        }
        cout <<"Your number is: " <<num <<endl;
    }
    cout <<"\033[32m Program is Ended! \033[0m";

    return 0;
}
```